
DEALER BEST PRACTICE SERIES

Improved Operations Using Specialized Inspection Facility

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November 5, 2010
1002-1.03-1203

1.0 Introduction

Performance and maintenance are key issues that are affected by communication, monitoring, and accountability. In the age of communication and technology a focus on specific problems and issues can be developed to enhance machine performance and maintenance. This can be done in a way that is understandable and attainable across the organization.

One can see this method of execution with the Albion Sands' Mobile Maintenance Practice focused on improving Physical Availability. As the process exists now, Pre-PMs are conducted 5 days prior to major service; this practice entails bringing the intended truck into the service shop for inspection and S-O-S™. However, this did not maximize machine downtime aligned with Albion's Maintenance Strategy. Reliability and inefficient downtime indicated that further investigation needed to occur.

2.0 Best Practice Description

The performance metrics in use at Albion today utilizes time as an essential part of the calculation. Analysis of shutdowns and stoppages are required when attempting to understand what is affecting physical availability. The following is the site definition of a Shutdown/Stoppage:

“Shutdown / Stoppage: An event that take a machine out of service
Shutdowns may be scheduled or unscheduled and include all types of
maintenance and repair activities excluding daily lubes, refueling and
inspections executed during lube or refueling activities.”

Observational analysis was completed with machine fueling, lubrication, and operator break time. Utilizing this time for inspections are considered opportune down time and do not count against MTTR, MTBS and hence Physical Availability.

A Daily Inspection Guide and a modified Pre-PM Program were developed to maximize the opportune downtime.

A custom facility was built for personnel to maximize execution during these short windows. The requirements of this facility included an office space, staging area for the Pre-Pm, and a small parts facility to execute minor repairs that could be completed during the daily inspection (Headlights, clamps, etc).

* Appendix A gives detailed layout and illustrations of proper layout.

3.0 Implementation Steps

- Process mapping and analysis of opportune downtime and with the mine site.
- Develop downtime best practices with current activates and to-be activities.
- Develop process and check sheets for the plan
- Develop plan for the facility depending on the level of work required

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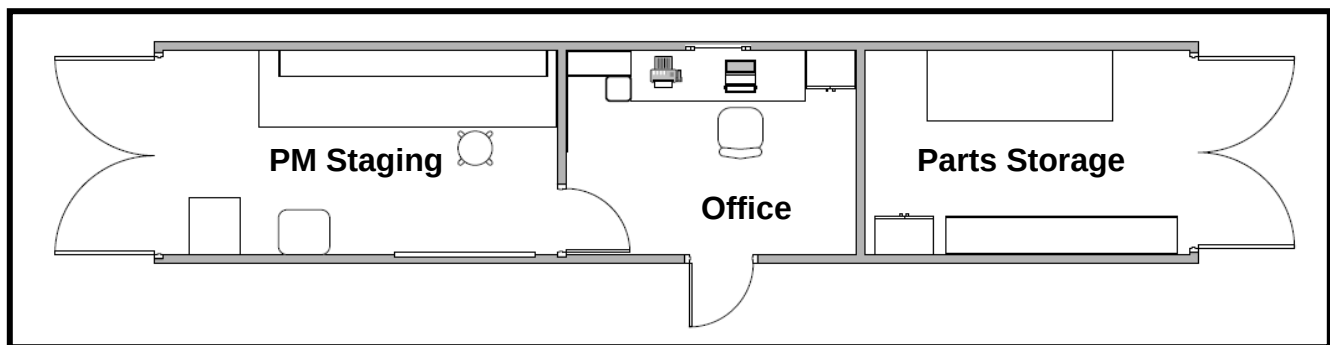
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For this site the best layout was determined to be a 12.9M (40 ft) Sea Container divided up into three sections. At one end located the PM staging area, the middle for the office, and the other end for parts storage as seen in the image below:



Resources necessary within the facility would be a 10 day plan, Daily Inspection Forms, Parts Order Forms, Pre-PM Forms, Time Card Books, and Work Order Request Forms. These resources can be stored on paper records, or on a computer system as to optimize communication, asset utilization, and proper tracking.

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A centralized location of the facility near the fuel and lubrication station was seen to be the best fit. This would provide technicians an optimal location and necessary resources. Location on the image below demonstrates layout in comparison to the haul roads.



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4.0 Benefits

At Albion the team was able to eliminate Pre-PM downtime by utilizing time during breaks, lubrication, and shift changes. This resulted in \$6,000,000 per 1.0%, and for the fleet of 797s they were able to save \$4,794,521 on a yearly basis (based on .8%).

Pre-PM Impact	
Trucks	38
Pre- PM Hr	5
Frequency	500
Yearly Hrs	7000
Number of PM	14
Downtime	2660
Un Availability	0.80%
Savings	\$4,794,521

The Daily inspections are allowing Albion to capture critical Backlogs that were not being captured. Minor repairs that can be completed during fueling are also completed as needed at the correct item.

5.0 Manpower Resources Required

- Dedicated support from both organizations on the supporting team members and execution
- Construction team for facility development and placement

6.0 Acknowledgements

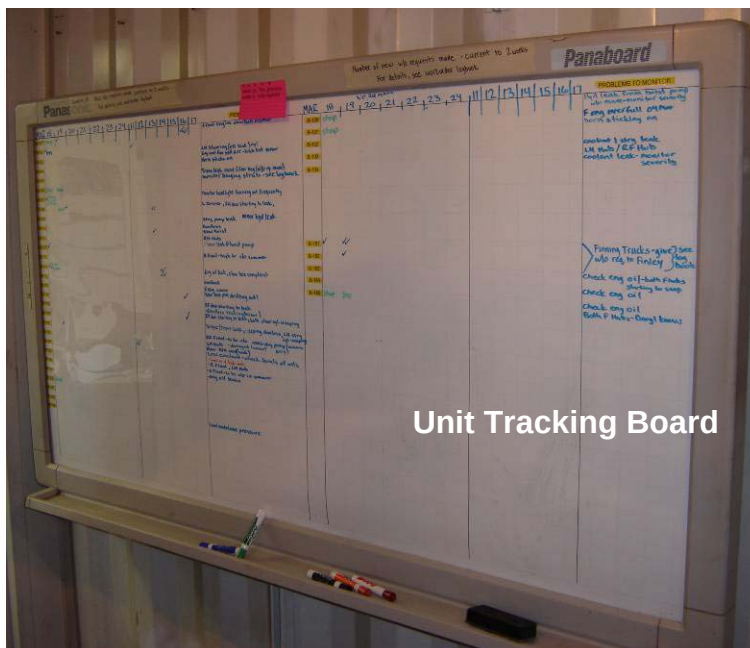
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Appendix:

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